



Support Automation — The Three Agents

Autonomous, unattended loops · each runs offline with no API key · Claude drop-in for production

SKILLS · AGENTS
NEITHER (plumbing)

Three agents take over manual support work. Each pairs **deterministic plumbing** (safety gates, data joins, sampling) with **swappable model judgement** — a mock reasoner runs here, a `*_claude.py` drop-in runs in production. The safety-critical logic never depends on the model.

1 Ticket Triage & Routing

AGENT + DETERMINISTIC GATE

Reads an inbox and, per ticket, runs a **deterministic escalation gate** for outage / security / legal **before** the model — these must fire 100% of the time. A swappable reasoner then judges category, team and urgency for the normal case. The gate wins on safety: it can only ratchet urgency **up**, never silently down, so a misclassifying model can never bury an emergency.

`python3 agents/ticket-triage-agent/triage_agent.py` 10 tickets → 4 escalated (gate caught all 4)

2 At-Risk Retention Briefing

AGENT · 3 SYSTEMS

Pulls from three systems — **billing, usage, tickets** — computes churn signals deterministically (seat-utilisation drop, login decline, negative sentiment, downgrades), then synthesises a one-page retention brief with an overall risk level and call talking points. Money figures are **historical record only, never a promise**: any offer must be verified against current policy before the call.

`python3 agents/at-risk-briefing-agent/briefing_agent.py acme-corp` Acme Corp → HIGH risk · 8 signals

3 Weekly Quality Grading

SKILL INSIDE AGENT

On a weekly cadence, deterministically **samples** closed tickets and applies the reusable **ticket-quality-grading skill** to each — a weighted five-category rubric with auto-fail compliance gates. It then synthesises a cross-ticket report: score distribution, auto-fail rate, recurring failure-tag patterns, per-agent and per-category clustering, and coaching recommendations. The clearest **Skill-inside-Agent** example.

`python3 agents/weekly-grading-agent/weekly_grading_agent.py` sample 10/12 → mean 71.2% · 2 auto-fails · coach “dev”

DESIGN RULE **Skills are libraries, Agents are workers, Neither is plumbing.** Non-determinism on a safety path is a bug, so escalation gates, data joins and sampling stay deterministic and auditable; only the open-ended judgement (classify, synthesise, grade) goes to the model — swap the mock for Claude and the agent logic is unchanged.